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OCTOBER 15, 1956
SUPERSEDING
Fed. Spec. QQ–I–706a
November 26, 1941

FEDERAL SPECIFICATION

TINPLATE (HOT DIP AND ELECTROLYTIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers tinplate, mild steel sheet coated on both surfaces with tin.

1.2 Classification.

1.2.1 *Types, grades, classes, and finishes.*—Tinplate shall be furnished in the following types, grades, classes, and finishes, as specified (see 6.2):

Types:

Type I.—Standard or general purpose plate.
Type II.—Drawing quality.
Type III.—Hard.

Grades:

Grade 1.—Hot dip, primes only (95 percent minimum primes).
Grade 2.—Hot dip, primes with seconds arising.
Grade 3.—Hot dip, unassorted.
Grade 4.—Electrolytic, unassorted.

Classes.—Tin plate shall be classified according to the method of coating (hot dip or electrolytic) and the weight of coating as follows:

Hot dip:

Class A.—Common cokes.
Class A2.—Best cokes. . . .
Class B.—1A charcoal.
Class C.—2A charcoal.

Electrolytic:

Class 25
Class 50
Class 75
Class 100

Finishes:

Bright — Hot dip and electrolytic, melted.

Dull-matte—Electrolytic only.

1.2.2 *Base weight and class limitations.*—The application of base weight to specific classes of tinplate is as shown in table I.

TABLE I.—*Base weight and class limitations*

Base weight pounds per base box (lb./bb.)	Applicable classes
55 ------	A, A2; Nos. 25, 50, 75, and 100
60 ------	A, A2; Nos. 25, 50, and 75
65 ------	A, A2; Nos. 25, 50, and 75
70 ------	A, A2; Nos. 25, 50, and 75
75 ------	A, A2; Nos. 25, 50, and 75
80 ------	A, A2, B, C; Nos. 25, 50, and 75
85 ------	A, A2, B, C; Nos. 25, 50, and 75
90 ------	A, A2, B, C; Nos. 25, 50, and 75
95 ------	A, A2, B, C; Nos. 25, 50, and 75
100 ------	A, A2, B, C; Nos. 25, 50, and 75
107 ------	A, A2, B, C; Nos. 25, 50, and 75
112 ------	A, A2, B, C; Nos. 25, 50, and 75
135 ------	A, A2, B, C; Nos. 25, 50, 75, and 100

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2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications.—There are no other specifications applicable to this specification.

2.2 Standards.—The following standards of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standard:

No. 48—Tolerances for Steel and Iron Wrought Products.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL–STD–163 — Preparation of Steel Products for Domestic Shipment (Storage) and Overseas Shipment.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Base metal.—The base metal in tinplate shall be low carbon cold reduced steel.

3.2 Types, grades, coatings and finishes.

3.2.1 *Types.*

3.2.1.1 *Type I, standard or general purpose plate.*—Tinplate of standard or general purpose, type I, shall withstand bending flat upon itself across the direction of rolling and flat around one thickness of the tinplate along the direction of rolling without cracking of the base metal at the outside of the bent portion.

3.2.1.2 *Type II, drawing quality.*—Tinplate of drawing quality shall produce, within given limits, parts as specified. A blueprint, sample, or description of the part will be furnished by the Government, when required. When the end use is not specified, the tinplate shall withstand bending flat upon itself (180°bend) in any direction without cracking of the base metal at the outside of the bent portion.

3.2.1.3 *Type III, hard.*—Hard tinplate, type III, shall withstand being bent flat around one thickness of the tinplate across the direction of rolling, without cracking of the base metal at the outside of the bent portion.

3.2.2 *Grades.*

3.2.2.1 *Grade 1, hot dip, primes only (95 percent minimum primes).* — Each lot of grade 1 tinplate primes shall consist of not less than 95 percent of primes.

3.2.2.2 *Grade 2, hot dip, primes with seconds arising.*—Each lot of grade 2 tinplate shall consist of not less than 75 percent primes and the remainder of seconds. Primes and seconds shall be packaged separately.

3.2.2.3 *Grade 3, hot dip, unassorted.*—Each lot of grade 3 tinplate shall consist of primes and seconds. Primes and seconds shall not be packaged separately.

3.2.2.4 *Grade 4, electrolytic, unassorted.*—Plates having minor imperfections in moderate degree may be included.

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3.2.3 *Coating.*

3.2.3.1 *Tin coating.*—The tin coating shall be of commercially pure tin. The weight per base box of tin coating on tinplate shall conform to the requirements of tables II and III.

TABLE II.—*Hot dip tin coating weight test values*

Class designation	Minimum average coating weight test value
	Pounds per base box
Class A	0.85
Class A2	1.19
Class B	1.80
Class C	2.30

TABLE III.—*Electrolytic tin plate coating weights*

Class designation	Nominal coating weight	Minimum average coating weight test value
	Pound per base box	*Pound per base box*
Class 25	0.25	0.22
Class 50	0.50	0.47
Class 75	0.75	0.70
Class 100	1.00	0.90

3.2.4 *Finishes.*

3.2.4.1 *Bright finish.*—Hot dip tinplate shall be of bright finish. Electrolytic tinplate shall be of bright (melted) finish, unless dull-matte (unmelted) finish is specified.

3.2.4.2 *Dull-matte finish.*—Electrolytic tinplate shall be dull-matte (unmelted) finish only when specified (see 6.2).

3.3 **Dimensions.**—The material shall conform to the ordered dimensions and weight, and shall be within the following tolerances:

	Federal Standard No. 48 Identification
Weight	7a1[1]
Width and length (see 3.3.1)	
Out-of-square	7a3
Resquaring	7a4
Camber	7a5

[1] A package is a unit consisting of 56, 112, or 224 plates (see 6.3.2).

3.3.1 The width and length shall be not less than the ordered dimensions (see 6.3.4). The greater of the two specified dimensions shall be considered the length.

3.4 **Identification marking.**—Each unit load shall be marked with the item name, base weight, size, type, grade, class, and number of plates.

3.5 **Workmanship.**—Tinplate, hot dip and electrolytic, shall be as nearly uniform in thickness as is practicable, flat, clean, smooth and as thoroughly and evenly coated with tin on both sides as good mill practice will permit. The tinplate shall be free from deep scratches, seams, pits, and other serious defects which would affect the intended use. Primes and seconds are defined as follows:

Primes.—Tinplate that is free from defects readily observable by the unaided eye.

Seconds.—Tinplate that has minor imperfections to a moderate degree or extent. These defects are in the coating, in the base metal, or are defects incidental to the manufacturing processes.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Lot.**—Unless otherwise specified in the contract or order, a lot shall consist of all tinplate of the same type, grade, class, finish, size, and base weight submitted for inspection at one time.

4.2 **Inspection.**

4.2.1 *Visual, dimensional, and weight inspection.*—All material shall be subject to

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visual inspection by the Government inspector. The Government inspector shall measure a sufficient number of sheets to satisfy himself whether or not the material meets the requirements of this specification.

4.2.1.1 *Squareness and edgewise curvature (camber) inspection.*—The squareness and edgewise curvature in the material under inspection shall be determined by placing selected samples (see 4.2.1) on a level surface and measuring the out-of-squareness and the depth of arc in the test samples by means of a straight edge and a steel scale.

4.2.1.2 *Base box weight inspection.*—Individual tinplates, packages, or the complete lot shall be weighed to determine conformance with the requirements of 7a1 of Federal Standard No. 48.

4.3 Tests.

4.3.1 *Sampling for tin coating and bend test.*—One sheet of tinplate shall be taken at random by the Government inspector from each 50 packages or fraction thereof in the lot, with a minimum of three samples (see 3.3).

4.3.2 *Bend test.*

4.3.2.1 One transverse (types I and II only) and one longitudinal bend test specimen (all types) shall be cut or sheared from each of the samples representing a lot. The test specimens shall be approximately 2 by 6 inches and shall be identified in regard to rolling direction. No portion of the test specimen shall be closer than ½ inch from the edge of the sample sheet.

4.3.2.2 The test specimens shall be bent and flattened as specified in 3.2.1.

4.3.3 *Tin-coating weight test.*

4.3.3.1 From each plate selected for test by the Government inspector, three specimens shall be taken on a diagonal across hot dip plate or directly across the plating direction on electrolytic plate. Circular specimens 2.257 ± 0.010 inches in diameter or square specimens 2 by 2 inches ± 0.010 inch shall be used. In order to secure representative samples they shall be taken so that the outer edge of the sample is at least one inch from any edge of the plate.

4.3.3.2 The average value for the lot obtained from testing the selected test specimens shall represent the average tin coating weight per base box. The tin coating shall be determined by any recognized method such as the Bendix, X-ray, antimony chloride or Sellers. At the option of the Government inspector or in cases of dispute, the determination of the coating weight shall be made by the referee method (see 4.3.4) at a laboratory designated by the Government inspector.

4.3.3.3 *Preparation of test pieces.*—The test pieces shall be cleaned with chloroform, carbon tetrachloride, benzol, or gasoline, and finally with alcohol; dried thoroughly; then accurately weighed together.

4.3.4 Referee method for determination of tin coating weight.

4.3.4.1 *Apparatus.*

4.3.4.1.1 *Platinum contact device for stripping of sample.*—It is suggested that approximately 14-gage platinum wire at least 24 inches long be coiled into a double spiral whose outside diameter approximates that of the specimen. Approximately five inches of this wire is not coiled but is bent perpendicular to the spiral. The straight length of wire facilitates handling of the sample during the stripping operation.

4.3.4.1.2 *National Bureau of Standards certified burette.*—Fifty ml. capacity is preferred.

4.3.4.1.3 *Sellers apparatus.*—Sellers apparatus as described in Scott's Standard Methods of Chemical Analysis, Fifth Edition, page

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966, or any similar equipment may be used that will maintain an atmosphere of carbon dioxide in the flask.

4.3.4.1.4 *Reaction flask.*—Either a 500-ml. Erlenmeyer or Kjeldahl flask may be used.

4.3.4.2 *Reagents.*

4.3.4.2.1 *Aluminum wire.*—Relatively pure aluminum (not less than 99.50 percent). Seven inches of 12-gage wire weighs approximately one gram.

4.3.4.2.2 *Tin metal.*—99.95 percent tin or better (National Bureau of Standards melting point tin is approximately 99.99 percent pure).

4.3.4.2.3 *Starch solution (10 gm. per liter).*—Prepare by making a paste of 1.0 gm. of either soluble or arrowroot starch in about 5 ml. of water and addition to 100 ml. of boiling water. Cool before using. If a preservative is added, it will not be necessary to prepare this solution fresh daily.

4.3.4.2.4 *Air-free water.*—To 3 liters of distilled water, which has been boiled and cooled, add 30 gm. of sodium bicarbonate and 25 ml. of concentrated hydrochloric acid.

4.3.4.2.5 *Standard tin solution (1 ml. equals approximately 0.0029 gm. of tin).*—For standardizing the iodate solution it is preferable to take aliquots from a standard master solution rather than weigh individual tin samples. Prepare the standard tin solution by taking approximately 2.9 grams of pure tin and weighing it accurately. Dissolve the tin in 100 ml. of concentrated hydrochloric acid and then dilute to 1,000 ml., in a volumetric flask, with 1:1 hydrochloric acid (see notes 3 and 7).

4.3.4.2.6 *Standard potassium iodate solution (1 ml. equals approximately 0.0029 gm. of tin).*—Dissolve 1.7600 gm. of potassium iodate (AR)[1] and 15.0 gm. of potassium iodide (AR)[1]. When solution is complete, dilute to 1000 ml. in a volumetric flask with water. Standardize the iodate solution against standard tin solution using such volume of tin solution so as to give approximately the same titration as the sample. This is essential. In standardizing the iodate solution, the standard tin solution must be carried through all steps indicated in the "Procedure." No provision is made for the deduction of a blank since this cancels out when the titrations for the sample and standard solutions are approximately equivalent. In some cases it may be necessary to run a series of standards containing various amounts of tin to correspond to samples of varying tin content (see notes 1 and 2).

[1] Analytical reagent.

4.3.4.3 *Procedure.*

4.3.4.3.1 A 4-square-inch tinplate specimen is placed between the spirals of the platinum contact device and immersed in 100 ml. of hydrochloric acid (1:1) contained in 250-ml. beaker or any suitable container. When the coating, including the alloy layer, is removed, the specimen is withdrawn from the solution and any adhering solution washed into the original beaker with three portions of distilled water. The contents of the beaker are then quantitatively transferred to a 500-ml. Erlenmeyer or Kjeldahl flask, and 1 gm. of aluminum wire is added. The flask is connected to the Seller's apparatus. After the aluminum is in solution, heat is applied and the solution gently boiled for approximately fifteen minutes. Remove the heat and cool with tap water to approximately 20°C. Remove the flask from the Sellar's apparatus, immediately add two or three marble chips and wash the sides of the flask with 60 ml. of air-free water containing 5 ml. of starch solution. Titrate immediately with standard iodate solution to a blue end point.

4.3.4.4 *Calculation.*—(ml. of standard iodate) (tin factor) (17.28) = pound of tin per base box (see 4.3.4.5).

4.3.4.5 *Notes.*

(1) The standard iodate described under "Reagents" will give suitable titrant volumes

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when used on tinplate from 0.50 to 1.50 pounds of tin per base box. For determining tin coating weights outside of this range an iodate solution of such a concentration should be prepared so that suitable titrant volumes are obtained (that is, iodate solution should be stronger or weaker than that indicated).

(2) It is suggested that it may be preferable to analyze the unknown samples prior to standardizing the iodate solution so that suitable volumes of standard tin solution can be selected. Such a procedure will result in more accurate standardization of iodate solutions and at the same time eliminate unnecessary standardizations.

(3) In making the standard tin solution up to volume, care must be employed in maintaining the temperature at which the volumetric flask was calibrated. This applies when measuring aliquots for standardization purposes. All standard solutions should be measured with a standard burette.

(4) For stripping hot-dipped tinplate, the coating is removed more rapidly by using 90 percent hydrochloric acid by volume and warming to approximately 70°C.

(5) The customary practice of allowing burettes to drain before reading the volume must be employed.

(6) All burette readings must be corrected in accordance with the National Bureau of Standards certificate.

(7) If a National Bureau of Standards certified volumetric flask is not available, the flask used should be standardized with a standard burette.

4.4 Rejection.—If the tinplate fails to meet any of the requirements of this specification, the lot represented by the specimens or samples shall be subject to rejection.

4.5 Retest.—A lot rejected for failing to meet the dimensional, squareness, camber, grading, coating weight or workmanship requirements may be resubmitted for Government inspection provided that the manufacturer, after having been informed of the reasons for rejection, has inspected each unit in the lot for the deficiency noted and has removed or reworked all non-conforming units. One reinspection only will be permitted.

5. PREPARATION FOR DELIVERY

5.1 Nonmilitary agencies.

5.1.1 *Packaging.*—Tinplate shall be packaged according to type, grade, class, base weight, and size.

5.1.1.1 Primes and seconds in grade 2 shall be packaged separately. Primes and seconds in grade 3 shall not be packaged separately.

5.1.1.2 Charcoal tinplate, except 1A charcoal (class B), shall be packaged with a tissue paper between successive sheets. 1A charcoal tinplate shall be packaged with interposed tissue papers when specified (see 6.2).

5.1.2 *Packing.* — The material shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose in such a manner as to insure acceptance and safe delivery to the designated point by common or other carrier at the lowest rate of the carrier.

5.1.3 *Marking.*—Shipping containers shall be marked with the name of the item, base weight, size, class, type, grade, and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

5.2 Military agencies.

5.2.1 *Packaging and packing.*

5.2.1.1 *Levels A and B.*—Tinplate shall be prepared for the level of shipment as specified in accordance with Military Standard MIL–STD–163. In addition, the requirements of 5.1.1. shall apply.

5.2.1.2 *Level C.*—Tinplate shall be packaged and packed in accordance with 5.1.1 and 5.1.2.

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5.2.2 *Marking.*—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–163.

6. NOTES

6.1 Intended use.—Tinplate covered by this specification is intended for use where a neat appearance, protective properties, and easy solderability are of prime importance as, for example, in the manufacture of cans, containers, houseware, kitchenware, etc. Tinplate is commonly used as containers for food. Tinplate is not suitable for gasoline tanks, since it corrodes under these conditions. Terneplate is commonly used for this purpose.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein and should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, grade, class, finish, base weight, quantity, and net weight (see 1.2).

(*c*) When dull-matte finish is required for electrolytic tinplate (see 3.2.4).

(*d*) Dimensions (see 3.3).

(*e*) When interposed tissue papers are required for packaging class B tinplate (see 5.1.1.2).

(*f*) Level of packaging and packing required (see 5.2).

6.2.1 Information on the intended end use should be given to enable the producer to select the proper combination of processing treatments.

6.3 Sheets.

6.3.1 Tinplate in sheets is the common form in which tinplate is commercially available. The sizes range in steps of $\frac{1}{16}$ inch from 10 to 32 inches in width and in steps of $\frac{1}{16}$ inch from 14 to 48 inches in length.

6.3.2 The quantity contained in a package is traditional and was originally set at 112 to conform to the English weight system and to permit manual handling of packages conveniently and individually. Units of 56 and 224 are used for smaller and larger than normal sizes to keep the package weight within the range of convenient manual handling

6.3.3 In modern practice tinplate is packed in bundles which are equivalent to 10, 12, or 15 packages. The package is not ordinarily used as a shipping unit but is essentially a unit of quantity.

6.3.4 The practice in the production of tinplate is to shear to approximately $\frac{1}{8}$ inch in width and approximately $\frac{1}{4}$ inch in length over the specified dimensions.

6.4 Electrolytic tinplate is also available in limited quantity in coil form, subject to negotiation between the purchaser and supplier. Because of the nature of coils, individual areas or portions of the coils will have defects present to a greater extent than may occur in cut sizes.

6.5 Base box weights.—Tinplate is selected and sold on the basis of its weight per unit area rather than on a gage or thickness basis. The unit of area is the base box, equal to the area of 112 sheets, 14 by 20 inches, or 31,360 square inches (217.8 square feet). Tables I and IV list the base box weights in pounds of tinplate commonly produced, the equivalent weight per square foot, the approximate thickness in inches, and old trade symbols if any were used. The approximate thicknesses and old trade symbols are included for information only and should not be used in the invitation for bids, or contracts or orders to specify the required tin plate.

6.6 Since a base box is equivalent in area to 112 sheets of 280 square inches each, for other sizes the number of base boxes in a package of 112 plates may be determined from the following formula:

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$$\text{Number of base boxes in package} = \frac{A}{280}$$

Where A = the ordered size in square inches for an individual plate.

Ratio tables which provide the same information are common in the tinplate industry.

6.7 Transportation description.—The applicable transportation description and carload and truckload minimum weights are as follows:

Tinplate, plain.
Carload weight 40,000 pounds.
Truckload weight 36,000 pounds.

6.8 Classes A, A_2, B and C of Federal Specification QQ–I–706a are essentially equivalent to and interchangeable with the same classes of this specification for the same type, grade, and finish. Classes A_3, D, E, F, and G are discontinued. When class A_3 is specified on drawings, etc., class A_2 should be procured.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—O
Navy—ShOr
Air Force.

TABLE IV.—*Tinplate base weights, approximate thicknesses, and old trade symbols. For information purposes only*

Weight per base box (base box designation)	Equivalent weight per square foot	Approximate thickness[1]	Old trade symbol
Pounds	*Pounds*	*Inch*	
55	0.253	0.0061	---
60	.276	.0066	---
65	.299	.0072	---
70	.321	.0077	---
75	.344	.0083	---
80	.367	.0088	---
85	.390	.0094	---
90	.413	.0099	---
95	.436	.0105	---
100	.459	.0110	ICL
107	.491	.0118	IC
112	.514	.0123	---
118	.542	.0130	---
128	.588	.0141	IXL
135	.620	.0149	IX
139	.638	.0153	DC
148	.680	.0163	2XL
155	.712	.0171	2X
168	.771	.0185	3XL
175	.804	.0193	3X
180	.827	.0198	DX
188	.863	.0207	4XL
195	.895	.0215	4X
208	.955	.0229	5XL
210	.964	.0231	D2X
215	.987	.0237	5X
228	1.047	.0251	6XL
235	1.079	.0259	6X
240	1.102	.0264	D3X
248	1.139	.0273	7XL
255	1.171	.0281	7X
268	1.231	.0295	8XL
270	1.240	.0297	D4X
275	1.263	.0303	8X

[1] When ordering, tinplate should be specified by weight per base box.

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